

# ROBOTIC KITS

## Explore Your Creativity

**Put your creativity and imagination to test with a robotic kit. There are lot many kits out there for beginners to advanced experimenters. Choose one according to your preference and the complexity you want**



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**R**obotic kits are a lot of fun. These let you assemble your own robot and see it work.

Many different types of robotic kits are available in India, ranging from basic robot kits to programmable kits (like kits for wheel bots, hexapods, bipeds and redundant robotic manipulators along with software toolkit). These meet the needs right from school level up to

postgraduate and research level.

Among basic types of kits, most are either remote-controlled or basic autonomous robots.

Swarm robots and legged robots, on the other hand, are capable of fully autonomous control. These robots can interact with each other and do a particular task in a swarm rather than individually. Higher-end robots like humanoids are still not easily available in India as demand is very less.

There are the options of ready-to-use kits and do-it-yourself kits. Complete kits include everything to build a robot, such as circuit board, sensors, motors, wheels, battery and programmer. The second option is individual parts. Here you can select your own electronic platforms, sensors, power source and other mechanical parts. Thus you can customise the robot as per your need.

"The most common and popular robotic kits are programmable robots. These kits are mobile robots equipped with sensors, motors and processor which can perform different activities according to the program. They are made for all age groups (for school children as well as graduate students)," says Adwait Deshpande, R&D head, robotics, Thinklabs.

If you are contemplating to buy a robotic kit, choose one according to your need and the complexity you want.

### What do the kits consist of?

Robot kits come with chassis/body parts, servo or DC motors, wheels, muscle wire, power supply, batteries, heat-sinks, servo controllers, sensors, MCU-based control board, serial cables, USB cables and programming platform/software/sample codes.

"The chassis forms the base of the robot, the motors control the movements of wheels or other body parts, the interconnecting wires interface the modules to the main circuitry and the power supply unit supplies the power or charges the battery. The sensors help the robot to take decisions by verifying the parameters. The micro-controller stores the program for robot control, sensor reading, etc. The cables interface the MCU board to the PC to load different programs. The IDE should be installed in a PC to compile and load different programs," explains Jiss John Thomas, senior project leader, rhydoLABZ.com, Rhydo Technologies.

All the basic robots have UART (serial) connectivity to interface with a PC and other devices. Advanced robots come with USB, CAN, I2C, SPI and wireless connectivity options.

The hardware platform performs the following functions:

1. Take input from sensors
2. Give output to displays and motors/actuators
3. Take feedback from actuators
4. Execute the program